EPSR-test-ex25 epsrwtsx: interactive X-ray weights

Program(s): epsrwtsx | chained passes: 1 | verification: ALL IDENTICAL (1 pass)

What this case tests

Exercises `epsrwtsx`, the interactive counterpart of `xwts`: for every atom type the user types the X-ray form-factor symbol and the MAFF q/delta values, then the output type, normalisation and base name. The resulting `X-ray.wts` file supplies `epsr` with the form-factor weights of an X-ray data set [1,2].

Code exercised

Fortran (original, fortran_src/):

- `epsrwtsx.f`: `epsrwtsx` (prompts, `nrho` check, data type from `nrho`/`normx`/`iafformaff`, writing `<base>X-ray.wts`), `form_fact` (search of `f0_WaasKirf.dat`)
- `ato_read_and_save.f`: `minitc`; `file_and_directory_commands.f`: `get_full_filename`

Rust (port, rust/epsr-core/src/):

- `tools/epsrwtsx.rs`: `run`
- `tools/xwts.rs`: `form_fact` (shared with `xwts`); `ato.rs`: `minitc`

Physical system

The EPSR-test mixture of 200 DMSO and 400 water molecules (3200 atoms; types S, C, O, H, O1, H1) for X-ray scattering, using the Waasmaier-Kirfel analytic form factors f0(s) = sum a_i exp(-b_i s²) + c [3]. `case1`: neutral-atom factors, per atom, independent-atom form factors (IAFF). `case2`: valence and ionic factors (`Cval`, `O1-`, `O2-`, `Hh+`), MAFF parameters (e.g. 0.5/0.7 for the water oxygen), `normx 2` and `iafformaff 2`. `case3`: neutral factors per molecule.

Inputs

`200DMSO400Water.ato` and `f0_WaasKirf.dat` from EPSR-test; answers in `case{1,2,3}.stdin`. `case2` gives the invalid `nrho 3`, an unknown symbol `Xx` (re-asked, answered `O1-`), one MAFF pair separated by a comma (`0.5,0.7`) and a blank line before the base name `W2 more`; `case3` names the `.ato` file without extension.

How the accuracy was verified

`commands.txt` (3 program calls, run once) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; no masking except where noted below). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in 1 pass.

One pass of three `epsrwtsx` calls; compared files: `W1X-ray.wts`, `W2X-ray.wts`, `W3X-ray.wts`. Their first lines show the option handling: `1 1 1` for case 1, `-1 2 2` for case 2 (the invalid `nrho 3` becomes per molecule, which the Fortran signals as -1) and `-1 1 1` for case 3.

Notes

`f0_WaasKirf.dat` is read from the current working directory, as in the Fortran.

References

[1] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).

[2] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).

[3] D. Waasmaier and A. Kirfel, "New analytical scattering-factor functions for free atoms and ions", Acta Cryst. A51, 416-431 (1995).

[4] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.